

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-101A/B - Crude charge		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 1	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2 (double-suction 1st stage)
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Crude charge
5	Driver	Electric motor	6	Suction source	Crude tank (OSBL) via suction header
OPERATING CONDITIONS					
7	Liquid	Crude charge	8	Pumping temperature normal / design	30 / 60 C
9	Density at PT	859 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Estimated	12	Rated / normal flow	729.7 / 663.4 m3/h
13	Suction pressure (rated)	1.94 barg	14	Discharge pressure	23.94 barg
15	Differential pressure / head	22.0 bar / 261 m	16	NPSH available (est.)	26.7 m
17	NPSH required (est., vendor)	10.1 m	18	NPSH margin	16.6 m
19	Hydraulic power	445.9 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	2
23	Efficiency (est.)	72 %	24	Absorbed power (rated)	616.7 kW
25	Motor rating	710 kW	26	Motor voltage	4.16 kV, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	14" / 10" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Hydrodynamic / forced oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	12.9 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-102A/B - Desalted crude booster		DOC. No.	CFU-000-ME-DS-006
				REV / DATE	A / 2026-10-02
				STATUS	Issued for review (FEED)
				PREPARED	CLAUDE CODE CHK: - APP: -
				SHEET	2
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2 (double-suction 1st stage)
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Desalted crude booster
5	Driver	Electric motor	6	Suction source	D-101B desalted crude outlet
OPERATING CONDITIONS					
7	Liquid	Desalted crude booster	8	Pumping temperature normal / design	136 / 165 C
9	Density at PT	776 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Estimated	12	Rated / normal flow	805.4 / 732.2 m3/h
13	Suction pressure (rated)	9.94 barg	14	Discharge pressure	29.94 barg
15	Differential pressure / head	20.0 bar / 263 m	16	NPSH available (est.)	65.1 m
17	NPSH required (est., vendor)	10.8 m	18	NPSH margin	54.3 m
19	Hydraulic power	447.5 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	2
23	Efficiency (est.)	73 %	24	Absorbed power (rated)	615.9 kW
25	Motor rating	710 kW	26	Motor voltage	4.16 kV, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	16" / 12" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Hydrodynamic / forced oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	12.9 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI).					
2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-103A/B - Atm. reflux		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 3	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Atm. reflux
5	Driver	Electric motor	6	Suction source	D-102 HC compartment (bubble point)
OPERATING CONDITIONS					
7	Liquid	Atm. reflux	8	Pumping temperature normal / design	45 / 75 C
9	Density at PT	703 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	158.6 / 144.2 m3/h
13	Suction pressure (rated)	1.11 barg	14	Discharge pressure	8.11 barg
15	Differential pressure / head	7.0 bar / 102 m	16	NPSH available (est.)	6.1 m
17	NPSH required (est., vendor)	2.3 m	18	NPSH margin	3.8 m
19	Hydraulic power	30.8 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	1
23	Efficiency (est.)	67 %	24	Absorbed power (rated)	46.0 kW
25	Motor rating	55 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	6" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	1.0 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-104A/B - Unstabilised naphtha		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 4	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Unstabilised naphtha
5	Driver	Electric motor	6	Suction source	D-102 HC compartment (bubble point)
OPERATING CONDITIONS					
7	Liquid	Unstabilised naphtha	8	Pumping temperature normal / design	45 / 75 C
9	Density at PT	703 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	179.9 / 163.6 m3/h
13	Suction pressure (rated)	1.11 barg	14	Discharge pressure	17.11 barg
15	Differential pressure / head	16.0 bar / 232 m	16	NPSH available (est.)	6.1 m
17	NPSH required (est., vendor)	2.5 m	18	NPSH margin	3.6 m
19	Hydraulic power	80.0 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	2
23	Efficiency (est.)	67 %	24	Absorbed power (rated)	118.6 kW
25	Motor rating	160 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	8" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	2.2 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-105A/B - Overhead sour water		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 5	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Overhead sour water
5	Driver	Electric motor	6	Suction source	D-102 water boot
OPERATING CONDITIONS					
7	Liquid	Overhead sour water	8	Pumping temperature normal / design	45 / 75 C
9	Density at PT	990 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Water	12	Rated / normal flow	14.4 / 13.1 m3/h
13	Suction pressure (rated)	0.98 barg	14	Discharge pressure	5.98 barg
15	Differential pressure / head	5.0 bar / 51 m	16	NPSH available (est.)	19.5 m
17	NPSH required (est., vendor)	1.2 m	18	NPSH margin	18.3 m
19	Hydraulic power	2.0 kW	20	Corrosive / erosive	Sour
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	59 %	24	Absorbed power (rated)	3.4 kW
25	Motor rating	5.5 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	2" / 2" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.4 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-106A/B - TPA pumparound		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 6	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	TPA pumparound
5	Driver	Electric motor	6	Suction source	C-101 tray 3 draw sump
OPERATING CONDITIONS					
7	Liquid	TPA pumparound	8	Pumping temperature normal / design	153 / 185 C
9	Density at PT	627 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	292.6 / 266.0 m3/h
13	Suction pressure (rated)	3.56 barg	14	Discharge pressure	11.56 barg
15	Differential pressure / head	8.0 bar / 130 m	16	NPSH available (est.)	38.5 m
17	NPSH required (est., vendor)	8.7 m	18	NPSH margin	29.8 m
19	Hydraulic power	65.0 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	69 %	24	Absorbed power (rated)	94.0 kW
25	Motor rating	110 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	10" / 8" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	1.6 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-107A/B - MPA pumparound		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 7	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	MPA pumparound
5	Driver	Electric motor	6	Suction source	C-101 tray 13 draw sump
OPERATING CONDITIONS					
7	Liquid	MPA pumparound	8	Pumping temperature normal / design	240 / 270 C
9	Density at PT	592 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	387.4 / 352.2 m3/h
13	Suction pressure (rated)	3.14 barg	14	Discharge pressure	11.14 barg
15	Differential pressure / head	8.0 bar / 138 m	16	NPSH available (est.)	32.0 m
17	NPSH required (est., vendor)	4.2 m	18	NPSH margin	27.8 m
19	Hydraulic power	86.1 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	1
23	Efficiency (est.)	70 %	24	Absorbed power (rated)	122.8 kW
25	Motor rating	160 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	10" / 8" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	C-6 (12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	3.5 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-108A/B - BPA pumparound		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 8	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	BPA pumparound
5	Driver	Electric motor	6	Suction source	C-101 tray 25 draw sump
OPERATING CONDITIONS					
7	Liquid	BPA pumparound	8	Pumping temperature normal / design	314 / 345 C
9	Density at PT	573 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	317.9 / 289.0 m3/h
13	Suction pressure (rated)	2.75 barg	14	Discharge pressure	10.75 barg
15	Differential pressure / head	8.0 bar / 142 m	16	NPSH available (est.)	24.2 m
17	NPSH required (est., vendor)	3.7 m	18	NPSH margin	20.6 m
19	Hydraulic power	70.6 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	1
23	Efficiency (est.)	69 %	24	Absorbed power (rated)	101.8 kW
25	Motor rating	132 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	10" / 8" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	A-8 (316)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	3.0 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-109A/B - Kero product		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 9			
GENERAL							
1	Applicable standard	API 610 12th ed. / ISO 13709		2	Pump type	BB2	
3	Number of pumps	2 (1 operating + 1 spare)		4	Service	Kero product	
5	Driver	Electric motor		6	Suction source	C-102 bottoms	
OPERATING CONDITIONS							
7	Liquid	Kero product		8	Pumping temperature normal / design	212 / 240 C	
9	Density at PT	621 kg/m3		10	Viscosity	TBD (process)	
11	Vapour pressure	Bubble point (= surface P)		12	Rated / normal flow	121.5 / 110.5 m3/h	
13	Suction pressure (rated)	1.48 barg		14	Discharge pressure	10.48 barg	
15	Differential pressure / head	9.0 bar / 148 m		16	NPSH available (est.)	3.1 m	
17	NPSH required (est., vendor)	1.9 m		18	NPSH margin	1.2 m	
19	Hydraulic power	30.4 kW		20	Corrosive / erosive	Sour (H2S) / naphthenic	
PERFORMANCE / CONSTRUCTION							
21	Speed (est.)	1780 rpm		22	Stages	1	
23	Efficiency (est.)	66 %		24	Absorbed power (rated)	46.0 kW	
25	Motor rating	55 kW		26	Motor voltage	480 V, 3 ph, 60 Hz	
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3		28	Suction / discharge nozzle	6" / 6" 300# RF (vendor)	
29	Casing MAWP	40 barg (incl. shut-off, TBD)		30	Material class (API 610 Annex H)	S-6 (CS/12Cr)	
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1		32	Bearings / lubrication	Rolling / ring oil	
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted		34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)	
35	Weight pump+motor+base (est.)	1.7 t each					
NOTES							
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.							

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-110A/B - Diesel product		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 10	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Diesel product
5	Driver	Electric motor	6	Suction source	C-103 bottoms
OPERATING CONDITIONS					
7	Liquid	Diesel product	8	Pumping temperature normal / design	294 / 325 C
9	Density at PT	594 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	151.0 / 137.3 m3/h
13	Suction pressure (rated)	1.60 barg	14	Discharge pressure	10.60 barg
15	Differential pressure / head	9.0 bar / 154 m	16	NPSH available (est.)	3.6 m
17	NPSH required (est., vendor)	2.2 m	18	NPSH margin	1.4 m
19	Hydraulic power	37.8 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	2
23	Efficiency (est.)	67 %	24	Absorbed power (rated)	56.5 kW
25	Motor rating	75 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	6" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	C-6 (12Cr)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	2.1 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-111A/B - Ago product		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 11	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Ago product
5	Driver	Electric motor	6	Suction source	C-104 bottoms
OPERATING CONDITIONS					
7	Liquid	Ago product	8	Pumping temperature normal / design	335 / 365 C
9	Density at PT	601 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	83.8 / 76.2 m3/h
13	Suction pressure (rated)	1.64 barg	14	Discharge pressure	10.64 barg
15	Differential pressure / head	9.0 bar / 153 m	16	NPSH available (est.)	2.6 m
17	NPSH required (est., vendor)	1.5 m	18	NPSH margin	1.1 m
19	Hydraulic power	21.0 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	2
23	Efficiency (est.)	65 %	24	Absorbed power (rated)	32.4 kW
25	Motor rating	45 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	6" / 4" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	A-8 (316)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	1.6 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-112A/B - Atm. residue / vacuum heater charge		DOC. No.	CFU-000-ME-DS-006
				REV / DATE	A / 2026-10-02
				STATUS	Issued for review (FEED)
				PREPARED	CLAUDE CODE CHK: - APP: -
				SHEET	12
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Atm. residue / vacuum heater charge
5	Driver	Electric motor	6	Suction source	C-101 bottoms
OPERATING CONDITIONS					
7	Liquid	Atm. residue / vacuum heater charge	8	Pumping temperature normal / design	349 / 380 C
9	Density at PT	710 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	397.8 / 361.6 m3/h
13	Suction pressure (rated)	1.86 barg	14	Discharge pressure	18.86 barg
15	Differential pressure / head	17.0 bar / 244 m	16	NPSH available (est.)	5.6 m
17	NPSH required (est., vendor)	4.2 m	18	NPSH margin	1.4 m
19	Hydraulic power	187.8 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	2
23	Efficiency (est.)	70 %	24	Absorbed power (rated)	267.5 kW
25	Motor rating	315 kW	26	Motor voltage	4.16 kV, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	10" / 8" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	A-8 (316)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	6.2 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI).					
2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					
3. HOLD: differential pressure inconsistent with H&MB stream 19 (see CAL-001 S13).					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-114A/B - Desalter wash water		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 13	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Desalter wash water
5	Driver	Electric motor	6	Suction source	Stripped sour water / utility water (OSBL) at 1.0 bar~
OPERATING CONDITIONS					
7	Liquid	Desalter wash water	8	Pumping temperature normal / design	50 / 80 C
9	Density at PT	990 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Estimated	12	Rated / normal flow	36.8 / 33.4 m3/h
13	Suction pressure (rated)	0.93 barg	14	Discharge pressure	8.93 barg
15	Differential pressure / head	8.0 bar / 82 m	16	NPSH available (est.)	18.8 m
17	NPSH required (est., vendor)	2.2 m	18	NPSH margin	16.6 m
19	Hydraulic power	8.2 kW	20	Corrosive / erosive	Sour
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	62 %	24	Absorbed power (rated)	13.2 kW
25	Motor rating	18.5 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	3" / 3" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.5 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-115A/B - Stabiliser reflux / LPG		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 14	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Stabiliser reflux / LPG
5	Driver	Electric motor	6	Suction source	D-105 (bubble point)
OPERATING CONDITIONS					
7	Liquid	Stabiliser reflux / LPG	8	Pumping temperature normal / design	45 / 75 C
9	Density at PT	530 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	38.8 / 35.3 m3/h
13	Suction pressure (rated)	10.69 barg	14	Discharge pressure	18.69 barg
15	Differential pressure / head	8.0 bar / 154 m	16	NPSH available (est.)	3.9 m
17	NPSH required (est., vendor)	2.3 m	18	NPSH margin	1.6 m
19	Hydraulic power	8.6 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	62 %	24	Absorbed power (rated)	13.9 kW
25	Motor rating	18.5 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	3" / 3" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.5 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-116A/B - Splitter reflux / LN		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 15	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Splitter reflux / LN
5	Driver	Electric motor	6	Suction source	D-106 (bubble point)
OPERATING CONDITIONS					
7	Liquid	Splitter reflux / LN	8	Pumping temperature normal / design	50 / 80 C
9	Density at PT	640 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	146.3 / 133.0 m3/h
13	Suction pressure (rated)	1.10 barg	14	Discharge pressure	8.10 barg
15	Differential pressure / head	7.0 bar / 111 m	16	NPSH available (est.)	5.0 m
17	NPSH required (est., vendor)	2.2 m	18	NPSH margin	2.8 m
19	Hydraulic power	28.4 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	1
23	Efficiency (est.)	67 %	24	Absorbed power (rated)	42.6 kW
25	Motor rating	55 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	6" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	1.0 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-117A/B - Heavy naphtha product		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 16	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Heavy naphtha product
5	Driver	Electric motor	6	Suction source	C-106 bottoms
OPERATING CONDITIONS					
7	Liquid	Heavy naphtha product	8	Pumping temperature normal / design	152 / 185 C
9	Density at PT	624 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	142.5 / 129.5 m3/h
13	Suction pressure (rated)	1.89 barg	14	Discharge pressure	7.89 barg
15	Differential pressure / head	6.0 bar / 98 m	16	NPSH available (est.)	6.6 m
17	NPSH required (est., vendor)	5.4 m	18	NPSH margin	1.2 m
19	Hydraulic power	23.7 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	67 %	24	Absorbed power (rated)	35.6 kW
25	Motor rating	45 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	6" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.8 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		PUMP DATASHEET API 610 12th ed. / API 682 P-118 - Desalter mud-wash / recycle		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 17	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	1 (no spare - see note)	4	Service	Desalter mud-wash / recycle
5	Driver	Electric motor	6	Suction source	D-101A/B brine (sub-cooled water at 11.5 barg)
OPERATING CONDITIONS					
7	Liquid	Desalter mud-wash / recycle	8	Pumping temperature normal / design	120 / 150 C
9	Density at PT	950 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Estimated	12	Rated / normal flow	19.2 / 17.4 m3/h
13	Suction pressure (rated)	11.43 barg	14	Discharge pressure	17.43 barg
15	Differential pressure / head	6.0 bar / 64 m	16	NPSH available (est.)	112.1 m
17	NPSH required (est., vendor)	1.4 m	18	NPSH margin	110.7 m
19	Hydraulic power	3.2 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	60 %	24	Absorbed power (rated)	5.4 kW
25	Motor rating	7.5 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	2" / 2" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.4 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PUMP DATASHEET API 610 12th ed. / API 682 P-201A/B - LVGO pumparound / product		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 18	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	LVGO pumparound / product
5	Driver	Electric motor	6	Suction source	C-201 LVGO collector
OPERATING CONDITIONS					
7	Liquid	LVGO pumparound / product	8	Pumping temperature normal / design	224 / 255 C
9	Density at PT	737 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	198.3 / 180.3 m3/h
13	Suction pressure (rated)	1.60 barg	14	Discharge pressure	11.60 barg
15	Differential pressure / head	10.0 bar / 138 m	16	NPSH available (est.)	35.7 m
17	NPSH required (est., vendor)	2.7 m	18	NPSH margin	33.0 m
19	Hydraulic power	55.1 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	1
23	Efficiency (est.)	68 %	24	Absorbed power (rated)	81.3 kW
25	Motor rating	110 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	8" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	2.7 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PUMP DATASHEET API 610 12th ed. / API 682 P-202A/B - HVGO pumparound / product		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 19	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	HVGO pumparound / product
5	Driver	Electric motor	6	Suction source	C-201 HVGO collector
OPERATING CONDITIONS					
7	Liquid	HVGO pumparound / product	8	Pumping temperature normal / design	315 / 345 C
9	Density at PT	709 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	391.0 / 355.4 m3/h
13	Suction pressure (rated)	0.98 barg	14	Discharge pressure	10.98 barg
15	Differential pressure / head	10.0 bar / 144 m	16	NPSH available (est.)	28.1 m
17	NPSH required (est., vendor)	4.2 m	18	NPSH margin	23.9 m
19	Hydraulic power	108.6 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	1
23	Efficiency (est.)	70 %	24	Absorbed power (rated)	154.8 kW
25	Motor rating	200 kW	26	Motor voltage	4.16 kV, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	10" / 8" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	A-8 (316)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	4.2 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PUMP DATASHEET API 610 12th ed. / API 682 P-203A/B - Slop wax		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 20	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Slop wax
5	Driver	Electric motor	6	Suction source	C-201 slop wax collector
OPERATING CONDITIONS					
7	Liquid	Slop wax	8	Pumping temperature normal / design	378 / 410 C
9	Density at PT	714 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	12.2 / 11.1 m3/h
13	Suction pressure (rated)	0.45 barg	14	Discharge pressure	8.45 barg
15	Differential pressure / head	8.0 bar / 114 m	16	NPSH available (est.)	20.0 m
17	NPSH required (est., vendor)	1.0 m	18	NPSH margin	19.0 m
19	Hydraulic power	2.7 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	58 %	24	Absorbed power (rated)	4.7 kW
25	Motor rating	5.5 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	2" / 2" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	A-8 (316)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.9 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PUMP DATASHEET API 610 12th ed. / API 682 P-204A/B - Vacuum residue (incl. quench)		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 21	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	BB2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Vacuum residue (incl. quench)
5	Driver	Electric motor	6	Suction source	C-201 boot (quenched VR)
OPERATING CONDITIONS					
7	Liquid	Vacuum residue (incl. quench)	8	Pumping temperature normal / design	366 / 395 C
9	Density at PT	761 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Bubble point (= surface P)	12	Rated / normal flow	195.6 / 177.8 m3/h
13	Suction pressure (rated)	-0.55 barg	14	Discharge pressure	13.45 barg
15	Differential pressure / head	14.0 bar / 187 m	16	NPSH available (est.)	5.2 m
17	NPSH required (est., vendor)	2.6 m	18	NPSH margin	2.6 m
19	Hydraulic power	76.1 kW	20	Corrosive / erosive	Sour (H2S) / naphthenic
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	1780 rpm	22	Stages	2
23	Efficiency (est.)	68 %	24	Absorbed power (rated)	112.3 kW
25	Motor rating	132 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	8" / 6" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	A-8 (316)
31	Mechanical seal	API 682 Plan 53B, API 682 Cat 2, Arr. 2	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	3.0 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PUMP DATASHEET API 610 12th ed. / API 682 P-205A/B - Hotwell sour water		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 22	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	OH2
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Hotwell sour water
5	Driver	Electric motor	6	Suction source	D-201 hotwell water compartment
OPERATING CONDITIONS					
7	Liquid	Hotwell sour water	8	Pumping temperature normal / design	45 / 75 C
9	Density at PT	990 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Water	12	Rated / normal flow	13.5 / 12.3 m3/h
13	Suction pressure (rated)	-0.02 barg	14	Discharge pressure	4.98 barg
15	Differential pressure / head	5.0 bar / 51 m	16	NPSH available (est.)	9.3 m
17	NPSH required (est., vendor)	1.1 m	18	NPSH margin	8.1 m
19	Hydraulic power	1.9 kW	20	Corrosive / erosive	Sour
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	58 %	24	Absorbed power (rated)	3.2 kW
25	Motor rating	3.7 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	2" / 2" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.3 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m.					

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		PUMP DATASHEET API 610 12th ed. / API 682 P-206A/B - Hotwell slop oil		DOC. No. CFU-000-ME-DS-006 REV / DATE A / 2026-10-02 STATUS Issued for review (FEED) PREPARED CLAUDE CODE CHK: - APP: - SHEET 23	
GENERAL					
1	Applicable standard	API 610 12th ed. / ISO 13709	2	Pump type	API 674/675 PD
3	Number of pumps	2 (1 operating + 1 spare)	4	Service	Hotwell slop oil
5	Driver	Electric motor	6	Suction source	D-201 slop oil compartment
OPERATING CONDITIONS					
7	Liquid	Hotwell slop oil	8	Pumping temperature normal / design	45 / 75 C
9	Density at PT	850 kg/m3	10	Viscosity	TBD (process)
11	Vapour pressure	Estimated	12	Rated / normal flow	0.6 / 0.6 m3/h
13	Suction pressure (rated)	-0.01 barg	14	Discharge pressure	4.99 barg
15	Differential pressure / head	5.0 bar / 60 m	16	NPSH available (est.)	11.4 m
17	NPSH required (est., vendor)	0.1 m	18	NPSH margin	11.3 m
19	Hydraulic power	0.1 kW	20	Corrosive / erosive	-
PERFORMANCE / CONSTRUCTION					
21	Speed (est.)	3560 rpm	22	Stages	1
23	Efficiency (est.)	55 %	24	Absorbed power (rated)	0.2 kW
25	Motor rating	0.75 kW	26	Motor voltage	480 V, 3 ph, 60 Hz
27	Motor enclosure / area	TEFC, Class I Div 2 Gr D / Zone 2 IIA T3	28	Suction / discharge nozzle	2" / 2" 300# RF (vendor)
29	Casing MAWP	40 barg (incl. shut-off, TBD)	30	Material class (API 610 Annex H)	S-6 (CS/12Cr)
31	Mechanical seal	API 682 Plan 11/52, API 682 Cat 2, Arr. 1	32	Bearings / lubrication	Rolling / ring oil
33	Coupling / baseplate	Flexible disc spacer / API 610 grouted	34	Minimum continuous flow	Vendor (min. flow bypass if < 30 % rated)
35	Weight pump+motor+base (est.)	0.3 t each			
NOTES					
1. Rated flow = 110 % normal (process). Heads per process sizing; hydraulic check by piping (CFU-PI). 2. NPSHr estimated with suction-specific speed 200 (metric, ~10,300 US); vendor to confirm NPSHr <= NPSHa - 1 m. 3. HOLD: flow below API 610 range - metering / PD pump recommended.					